

Supervised machine learning jointly predicts geographic location and climate of origin from genotypes

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Abstract

Introduction

Population genetic theory predicts that the distribution of genetic variation within and between populations is shaped by demographic history, spatial structure, and environmental pressures, and thus carries information about the processes that generate and maintain it (Lewontin, 1974; Hartl and Clark, 1980; Ellegren and Galtier, 2016). In spatially structured, locally adapted populations, genetic variation encodes information about both geographic location and climate of origin. Geographic structure arises because local mating and limited dispersal generate isolation by distance, causing alleles to be more similar among neighboring individuals than among those farther apart (Wright, 1943; Slatkin, 1993). Environmental selection, in contrast, produces allele frequency clines that parallel ecological gradients, as individuals carrying locally beneficial variants enjoy higher fitness in particular environments (Endler 1977; Kawecki and Ebert, 2004). At the same time, neutral demographic processes—historical range expansions, bottlenecks, and genetic drift—can generate allele frequency gradients that mimic adaptive signals (Excoffier, Foll, and Petit, 2009); Novembre and Di Rienzo 2009). Whereas neutral structure aligns primarily with geographic distance, adaptive signals track specific environmental variables—temperature, moisture, seasonality—that impose fitness differentials on individuals (Rellstab et al. 2015; Hoban et al., 2016; Tigano and Friesen, 2016; Coop et al., 2010).

Local adaptation provides the clearest link between genetic variation and environmental context. It arises when spatially heterogeneous selection outweighs the homogenizing effects of gene flow and drift, causing allele frequencies to covary with environmental gradients (Savolainen, Lascoux, and Merilä, 2013; Lenormand, 2002). This process not only increases fitness locally but

also sustains genetic diversity across heterogeneous landscapes by maintaining multiple adaptive alleles, ensuring that populations retain the variation on which selection can act as environments change (Hereford 2009). Classic models—Wright’s isolation-by-distance and stepping-stone frameworks (Wright, 1943), Levene’s multiple-niche equilibrium (Levene, 1953), and Endler’s cline theory (Endler 1977)—describe how selection, migration, and drift interact to shape predictable spatial patterns of variation. More recent syntheses show that even modest selection can create steep genomic clines under limited dispersal (Tigano and Friesen, 2016; Yeaman and Whitlock, 2011), while strong gene flow can swamp adaptive alleles unless they are tightly linked or of large effect (Slatkin, 1975). Together, these insights demonstrate that evolutionary forces structure spatial genetic variation in predictable ways, revealing how populations interact with their environments.

Understanding this relationship is especially urgent as anthropogenic climate change rapidly disrupts the match between populations and their environments (Calvin et al., 2023). As suitable geographic ranges shift, conservation managers face increasing pressure to anticipate where populations will persist under novel conditions. Classic approaches—common-garden experiments, reciprocal transplant trials, and provenance tests—have been indispensable for characterizing local adaptation, but they are slow, costly, and often limited in taxonomic or geographic scope (Aitken and Whitlock, 2013). Genomic tools offer a faster, more scalable alternative, and the emerging field of landscape genomics seeks to link spatial and environmental variation with patterns of genetic diversity (Manel et al., 2003; Rellstab et al., 2015). For forest species in particular, genomic approaches are increasingly informing seed transfer and assisted gene flow decisions (Aitken and Bemmels, 2016).

Genotype–environment association (GEA) methods have become central to this effort. These approaches identify putatively adaptive loci by testing for statistical associations between allele frequencies and environmental variables (Rellstab et al., 2015). Univariate methods such as Bayenv2 (Günther and Coop, 2013), latent factor mixed models (LFMM; Frichot et al., 2013), and BayPass (Gautier, 2015) test loci one at a time against environmental predictors, typically while controlling for population structure through covariance matrices or latent factors. These methods have been widely applied, but they are limited in their ability to capture polygenic adaptation involving many loci of small effect, and they can be confounded by neutral demographic processes that generate allele frequency gradients correlated with environmental variation (Lotterhos and Whitlock, 2015). Multivariate approaches address some of these limitations by analyzing many loci and environmental predictors simultaneously. Redundancy analysis (RDA) has proven effective at detecting multilocus adaptive signatures (Forester et al., 2018; Capblancq and Forester, 2021), and gradient forest methods model nonlinear allele–environment relationships across the genome (Fitzpatrick and Keller, 2015). More recently, the Weighted-Z Analysis (WZA) introduced a window-based framework for aggregating single-locus GEA statistics into region-level summaries of adaptive signal (Booker et al., 2024). Despite this progress, most GEA methods share key limitations: they test for associations with environment without jointly modeling geographic structure, and they do not produce direct predictions of an individual’s environment of origin from its genotype.

Machine learning (ML) offers a complementary approach that can overcome several of these constraints. ML methods excel at extracting patterns from high-dimensional data, naturally handling

nonlinear interactions among loci and correlated environmental predictors (Schrider and Andrew D. Kern, 2018). C. Battey, Peter L Ralph, and Andrew D Kern (2020) demonstrated that deep neural networks can predict the geographic origin of an individual from its genotype alone, using a tool called Locator. By learning from multilocus patterns of isolation by distance, Locator achieves high spatial accuracy, but it does not incorporate environmental covariates and therefore cannot distinguish adaptive signals from neutral spatial structure. Other recent work incorporates ecological context more explicitly: gradient forest methods link genomic variation to environmental predictors to forecast climate-driven maladaptation [Lotterhos et al., 2021], while multi-environment deep learning architectures in agricultural genomics integrate weather, soil, and other covariates to improve genotype performance prediction across climates [Washburn et al., 2021; Montesinos-López et al., 2022]. What remains missing is an approach that jointly models geographic and environmental prediction from genotypes, capturing both neutral spatial structure and adaptive environmental associations in a single framework.

Here we introduce EcoLocator, a deep neural network that extends the geographic prediction framework of Locator by jointly predicting geographic location and climate of origin from genomic data. Trained on georeferenced genotypes paired with historical climate data, the model learns multilocus associations between genetic variation and environmental gradients. When allele frequencies covary with climate variables, the model can infer climate of origin directly from genotype. By coupling spatial and environmental inference, EcoLocator can leverage signals of both isolation by distance and local adaptation. Downstream feature attribution methods such as SHAP (Lundberg and Lee, 2017) reveal how each input feature contributed to a given prediction, identifying which variants contribute most strongly to adaptive differentiation across landscapes.

We evaluate EcoLocator across a range of simulation parameterizations to identify the conditions that yield optimal performance, then extend the analysis to an empirical dataset of coastal Douglas-fir (*Pseudotsuga menziesii* var. *menziesii*). Simulations provide a controlled setting where evolutionary and environmental parameters are known, allowing us to benchmark prediction accuracy against ground truth before moving to more complex natural systems. As an empirical test case we focus on coastal Douglas-fir, a member of the Pine family that has its origins in the Paleocene (Hermann, 1985), and that has a natural range spanning extensive climatic gradients across the North American Pacific coast from California to British Columbia. While most populations show low genetic differentiation, high gene flow and high outcrossing rates (Slavov, Glenn T. Howe, and Adams, 2005), Douglas-fir populations still exhibit strong signatures of local adaptation, as adaptive quantitative phenotypes track climatic features like winter cold temperatures and summer aridity (J. Bradley St Clair, Mandel, and Vance-Borland, 2005). These features, combined with a long history of ecological and silvicultural study, make Douglas-fir a natural bridge between controlled simulations and real-world management applications. For the U.S. Forest Service, understanding seed source–climate relationships is central to understanding (mal)adaptation (J. Bradley St. Clair, Glenn T. Howe, and Kling, 2020), reforestation success and assisted migration planning (J. B. St Clair and G. T. Howe, 2007). By accurately matching genotypes to suitable environments, EcoLocator complements long-standing traditional knowledge on seed sourcing (such as ‘seed zones’ (Randall, 1996)), and expands management options that are essential for predicting individual tree

and forest contributions to productivity and resilience to climate change. Importantly, EcoLocator is not limited to this system; it offers a pathway to inform management in any species or taxon where empirical knowledge of local adaptation is scarce.

By integrating population genetic theory, simulation benchmarking, and empirical application, our study presents the first open-source approach for simultaneously inferring the geographic origin of a genotype and the climate to which it is adapted—precisely the information needed as shifting climates outpace traditional experimental approaches. These advances at the intersection of evolutionary biology and computational science open new avenues for understanding the genetic basis of adaptation, with broad implications for conservation and species management.

EcoLocator

EcoLocator is an open-source Python package for predicting the geographic origin and climate of origin of individual genotypes, accessible via both a command-line interface and a programmatic Python API.¹ Unlike traditional statistical methods that rely on fixed assumptions about allele–environment relationships, EcoLocator learns directly from high-dimensional genomic and environmental data. By jointly predicting location and environmental variables, it captures both spatial genetic structure and signals of local adaptation, providing a scalable, biologically informed approach for studying complex evolutionary patterns. Both the CLI and Python API accept standard genotype formats (VCF, Zarr, tab-delimited matrices), facilitating integration into existing genomic workflows whether scripting interactively or from the command line.

Preprocessing

EcoLocator converts genomic data from VCF, Zarr, and tab-delimited matrix formats into vectors of minor allele counts per individual using the scikit-allel (Miles et al., 2024) and NumPy (Oliphant and Team, 2024) libraries. Missing genotypes are imputed by drawing alleles from a binomial distribution parameterized by the site’s mean minor allele frequency, an approximation that preserves expected variation while minimizing bias. Sites can be filtered by a minor allele count threshold to reduce noise from rare variants that contribute little predictive signal. Geographic coordinates are normalized to zero mean and unit variance, and environmental covariates are standardized similarly prior to training.

Model architecture

EcoLocator employs a multilayer perceptron with a shared trunk and two task-specific output heads, trained under a supervised learning framework on labeled datasets of genotypes paired with known geographic coordinates and climate variables. Input genotypes pass through a batch normalization layer before entering the shared trunk, which consists of dense layers with ELU activations and a central dropout layer for regularization. The trunk branches into two heads — one predicting 2D

¹Source code, installation instructions, and vignettes are available at <https://github.com/kr-colab/ecolocator> and [VIGNETTEURL].

geographic coordinates, one predicting environmental covariates — each implemented as a small two-layer subnetwork. The network is optimized using the Adam optimizer with a custom multitask loss function (Equation 1), combining mean Euclidean distance for geographic prediction and MSE for environmental covariates, each scaled by an independently specified weight.

$$\mathcal{L} = w_{\text{loc}} \left(\frac{1}{n} \sum_{i=1}^n \sqrt{(\hat{x}_i - x_i)^2 + (\hat{y}_i - y_i)^2} \right) + w_{\text{env}} \left(\frac{1}{np} \sum_{i=1}^n \sum_{j=1}^p (\hat{z}_{ij} - z_{ij})^2 \right) \quad (1)$$

where n is the number of training samples, p is the number of environmental covariates, x_i and y_i are the true longitude and latitude of sample i , $\hat{x}_i$ and $\hat{y}_i$ are the predicted coordinates, z_{ij} is the true value of covariate j for sample i , $\hat{z}_{ij}$ is the predicted value, and $w_{\text{loc}}, w_{\text{env}} \geq 0$ are loss weights (both defaulting to 1.0, see **Supplemental Figure 1.**). The network is trained using a standard validation scheme in which a user-specified fraction of samples (default 0.9) is used for model fitting, while the remaining fraction is held out for validation, enabling convergence monitoring, early stopping, and adaptive learning rate reduction to prevent overfitting. By explicitly coupling geographic and climatic inference, the model captures both neutral spatial structure and environment-driven selection.

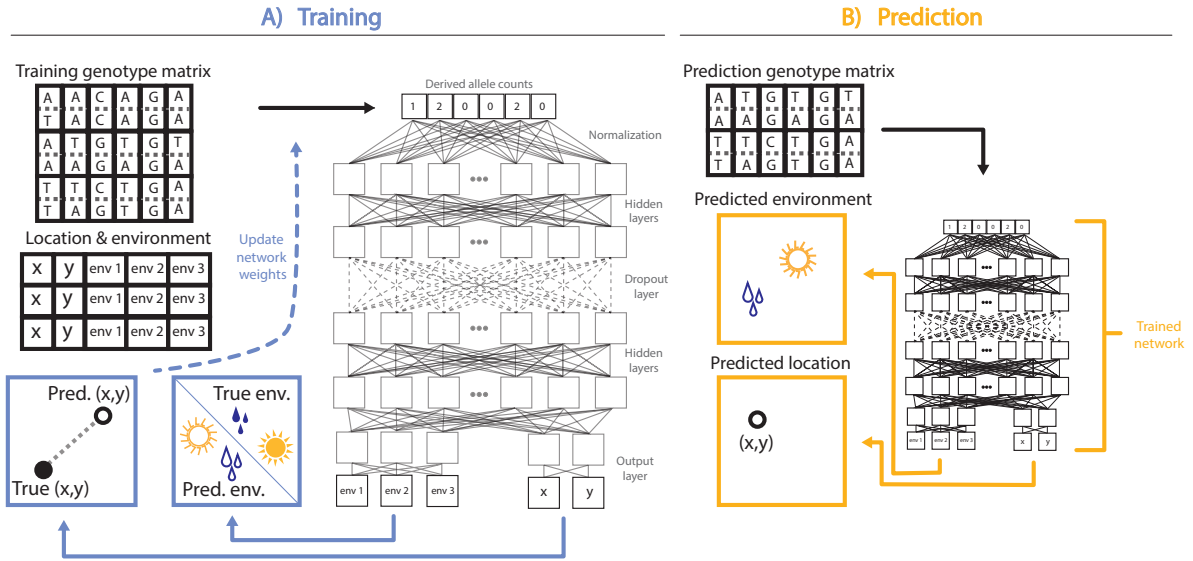


Figure 1: EcoLocator architecture. **Panel A** shows the training regime, with the inputs traversing the network in a supervised fashion; network weights are updated through backpropagation until predictions are optimized towards the true value, with the process shown here in blue. Once trained, EcoLocator enters the prediction step, shown here in **Panel B**. Given genotypes alone, the trained network jointly predicts climate-of-origin and geographic location, shown in yellow.

In practice, an end-to-end EcoLocator analysis consists of five steps:

1. Assemble a training dataset consisting of genotype data in a supported format (VCF, Zarr, or tab-delimited matrix) and a sample metadata file pairing each individual with its known geographic coordinates and environmental covariate values. Samples to be predicted should have their coordinates and covariates left blank.
2. Train the model using `ecolocator train` (or the Python API), specifying the genotype file, sample metadata, and an output directory. The model is fit to jointly predict geographic origin and environmental covariates, with early stopping and adaptive learning rate reduction applied automatically.
3. **(Optional)** Evaluate model performance using `ecolocator loo`, which iteratively holds out each sample with known coordinates, trains on the remainder, and predicts the held-out individual (leave-one-out cross validation). The resulting predictions can be compared against true values to assess spatial and environmental accuracy before applying the model to unknown samples.
4. Predict geographic origin and environmental covariates for samples with unknown locations using `ecolocator predict`, providing the saved model directory and a genotype file containing the target samples.
5. **(Optional)** Quantify per-SNP importance for predicted samples using `ecolocator attribute`, which applies SHAP values to identify loci driving each prediction.

The predicted coordinates and environmental values from steps 2–4 provide continuous estimates of geographic and climatic origin, and SHAP attributions from step 5 can be used to pinpoint the loci driving each prediction.

Methods

Simulating local adaptation in continuous space (non-Wright–Fisher)

To explore under which evolutionary scenarios EcoLocator performs best, we use forward-in-time population genetic simulations in SLiM v.4.2.2 (Haller and Messer, 2023). Starting with an initial population, forward-in-time simulation allows for the application of evolutionary forces such as mutation, selection, and genetic drift to shape a population through generations. By varying parameters such as dispersal, mutation rate, recombination rate, genetic architecture, and the standard deviation of the fitness function, the evolutionary trajectory of a given population can be changed in predictable ways. For example, in a simulation where a population experiences weak environmental selection and individuals in that population disperse far distances, it is expected that, due to the homogenizing effects of gene flow across the landscape, there would be little correlation between genetic variation and environmental variation (Lenormand, 2002; Kawecki and Ebert, 2004; Booker et al., 2024). This simulation setup allows systematic exploration of how ecological and evolutionary parameters, such as dispersal, selection strength, and population density, shape the phenotypic composition of populations throughout the landscape.

Fitness and reproduction

To capture the joint effects of selection, dispersal, and spatial dynamics, we simulate individuals in continuous space on top of an environmental grid. We initialize each simulation by randomly placing individuals across the landscape. The environment is represented as a 8×8 grid, where each cell has an environmental value that defines the local phenotypic optimum, θ . Values are generated with a Hurst exponent of zero ($H=0$), which removes spatial autocorrelation and creates a patchwork of optima. This setup allows us to isolate the effects of spatial dynamics and dispersal from those of environment-dependent selection.

Individuals experience fitness as a combination of stabilizing selection and density-dependent competition. Stabilizing selection follows a Gaussian function centered on the local optimum (Equation 2), where z is the individual's phenotype, θ is the optimum in the corresponding grid cell, and σ controls the strength of selection:

$$w(z, \theta) = \exp \left[-\frac{(z - \theta)^2}{2\sigma^2} \right] \quad (2)$$

Total fitness is then determined by multiplying this term with a density-dependent factor C_i , where competition is implemented as a Beverton–Holt style density regulation function, where the effective fitness of individual i is reduced according to the local Gaussian-weighted density of neighbors,

$$W_i = w(z_i, \theta_i) \cdot C_i \quad (3)$$

$$C_i = \frac{1}{1 + \rho D_i} \quad (4)$$

Thus, individuals in crowded regions are less likely to survive and reproduce. After survival, individuals disperse according to a user-specified kernel, which sets the spatial scale of gene flow. Dispersal redistributes individuals across the landscape, and the extent of this movement determines how much subpopulations mix across different environments in the landscape. Following dispersal, individuals mate with nearby partners, such that reproduction depends on both spatial proximity and local density. By varying the dispersal parameter and the standard deviation of the fitness function, we can impact the strength of local adaptation in the simulation (Figure 3).

Together, stabilizing selection, competition, dispersal, and local mating govern the survival and reproduction of individuals, creating a spatially explicit simulation framework that departs from Wright–Fisher assumptions. Local adaptation can then be quantified as the mean difference between each individual's phenotype and its local optimum, such that smaller values correspond to stronger local adaptation (Equation 5).

$$\text{Local Adaptation} = \frac{1}{N} \sum_{i=1}^N |P_i - O_i| \quad (5)$$

Genomic architecture and tree sequence recording

The genetics underlying adaptation are simulated with a 1 Mb genomic element carried by each individual. Mutations at quantitative trait loci (QTL) arise within this element, and their additive effects determine an individual’s phenotype. To explore whether the arrangement of adaptive loci influences the detectability of local adaptation, genomic architecture is varied by altering the distribution of QTLs: mutations may occur within a single 200 kb window in the center of the element, within three smaller 100 kb windows, or within five 100 kb windows distributed across the element. (Figure 3., panel c)

We only simulate non-neutral mutations within SLiM. Genomic information is stored in tree sequences (Haller, Galloway, et al., 2019), which record the ancestry and mutational history of all individuals. Neutral mutations are then overlaid onto these tree sequences using `msprime v.1.1.1` (Baumdicker et al., 2022), allowing a clear separation between adaptive genetic variation and neutral background variation. This compact representation provides the full genomic context needed to explore the link between genetic variation and environmental adaptation.

Varying dispersal, selection, and genetic architecture parameters

To evaluate the robustness of EcoLocator across diverse evolutionary contexts, we varied three parameters—dispersal, the strength of stabilizing selection, and the genomic architecture of adaptive loci. These axes of variation were chosen because they represent core processes shaping the detectability of local adaptation: gene flow homogenizes allele frequencies across space, selection drives divergence between environments, and genomic architecture determines how adaptive alleles are distributed across the genome. By exploring combinations of these parameters, we can assess under which evolutionary regimes spatial–environmental signals of adaptation are most readily captured.

Dispersal distance (σ_D) sets the spatial scale of gene flow in the simulation. Smaller values correspond to restricted movement, leading to stronger genetic differentiation among neighboring regions, whereas larger values represent more mixing across the landscape. We tested three dispersal regimes ($\sigma_D = 0.1, 0.3, 0.5$), which span cases where individuals remain close to their birthplace versus scenarios where they disperse broadly and homogenize allele frequencies.

The strength of stabilizing selection is governed by the standard deviation parameter (σ_K) in the Gaussian fitness function. Small values of σ_K impose strong penalties on phenotypes deviating from the local optimum, leading to sharper differentiation between environments, whereas large values correspond to weaker selection that tolerates a broader range of phenotypes. We evaluated three values ($\sigma_K = 0.3, 0.5, 0.7$) to capture strong, intermediate, and weak selection regimes.

To explore whether the genomic distribution of adaptive loci influences the detectability of local adaptation, we varied the arrangement of quantitative trait loci (QTLs) along the 1 Mb chromosome. In one scenario, all QTLs were clustered within a single 200 kb window at the center of the chromosome. In the second, QTLs were distributed across three smaller windows. In the third, they were spread across five windows distributed evenly along the chromosome. These treatments allowed us to test whether adaptive signals are easier to detect when concentrated in specific genomic regions

versus dispersed more broadly.

These parameter combinations generate a set of evolutionary scenarios that differ in the balance between gene flow, selection, and genomic distribution of adaptive alleles. In each parameter variation test case, we held the other parameters at an intermediate value (i.e., when testing dispersal parameters, we held σ_K at 0.5). While the units in SLiM are arbitrary, they can be interpreted qualitatively: smaller dispersal values correspond to localized neighborhoods, smaller (σ_K) values correspond to stronger local adaptation pressures, and greater clustering of QTLs corresponds to tighter genomic control of adaptive traits. By exploring this parameter space, we created simulations that range from highly structured, strongly adapted populations to less structured, weakly adapted populations. This variation provides the basis for evaluating when and where EcoLocator most effectively detects spatial–environmental signals of adaptation.

While these simulations allow us to systematically explore how key evolutionary parameters shape local adaptation in continuous space, they differ from the discrete, stepping-stone simulation framework used in prior evaluations of genome–environment association (GEA) methods (Booker et al., 2024). To facilitate more direct comparison with these approaches, we implemented an additional set of simulations based on the Wright–Fisher framework used to benchmark the Weighted-Z Analysis (WZA), a method that aggregates SNP-level association signals across genomic windows to detect loci involved in local adaptation.

Simulating local adaptation in discrete space (Wright–Fisher)

To complement our spatially explicit simulations, we implemented a second set of forward-in-time simulations based on the Wright–Fisher framework described in Booker et al. (2024). Simulation design and parameterization closely followed the implementation provided in the WZA GitHub repository (<https://github.com/TBooker/WZA/tree/master>), with minor modifications to ensure compatibility with the current version of SLiM. These simulations were designed to more closely mirror the evolutionary scenarios used to evaluate WZA, allowing for a more direct comparison between EcoLocator and existing GEA methods.

Following Booker et al. (2024), we simulated a two-dimensional stepping-stone metapopulation composed of discrete demes arranged on a grid, with a constant population size per deme under Wright–Fisher dynamics. Migration was restricted to neighboring demes, producing a pattern of isolation by distance and spatial genetic structure across the landscape. Environmental heterogeneity was defined using the British Columbia (BC) map of climatic variation, which assigns each deme a local phenotypic optimum based on discretized environmental values.

In contrast to our main simulations, which model stabilizing selection in continuous space, these Wright–Fisher simulations implement directional selection on a small number of causal loci. Specifically, a set of 12 loci distributed across the genome contribute to local adaptation, with alleles experiencing spatially varying fitness effects depending on the local environmental optimum. Selected mutations were modeled as semidominant, such that heterozygous individuals experience intermediate fitness effects, and the strength of selection was controlled by a fixed selection coefficient. This formulation produces strong genotype–environment associations localized to a relatively

small number of genomic regions.

Simulations were run in two phases: an initial neutral burn-in period to establish population structure and linkage patterns, followed by a directional selection phase during which adaptive alleles increase in frequency in a spatially heterogeneous manner. Tree-sequence recording was used to capture the full genealogical history of each simulation, and neutral mutations were subsequently overlaid to generate genome-wide variation.

To prepare as input to EcoLocator, deme coordinates were encoded as latitude and longitude, using the row and column indices of each deme in the grid, and the environmental variable was defined as the local phenotypic optimum for each deme. Genotype matrices were extracted from tree sequences and used as input for both EcoLocator and downstream GEA analyses.

These Wright–Fisher simulations provide a complementary scenario to our main simulations, in which local adaptation is driven by a smaller number of loci under directional selection and spatial structure is defined by discrete demes. As a result, they represent a comparatively simpler setting in which genotype–environment associations are strong and more localized in the genome. This allows us to assess how EcoLocator performs in a simpler regime, providing a useful contrast to more complex spatial scenarios.

Feature attribution

To quantify the contribution of individual genetic features to EcoLocator’s predictions, we applied SHapley Additive exPlanations (SHAP; Lundberg and Lee, 2017), a game-theoretic framework that assigns importance scores based on each feature’s marginal contribution to the model output across all possible feature coalitions. Attributions were computed separately for each prediction target (latitude, longitude, and environmental covariates), allowing the genetic contributions to be evaluated conditional on the response variable of interest. SHAP values were calculated on trained models using the genotype matrix as input, attributing predictive signal to individual loci while accounting for nonlinear interactions among sites. To complement SHAP-based attribution and assess the robustness of inferred signals, we performed a series of permutation-based sensitivity analyses in which subsets of the genotype matrix were shuffled across individuals prior to prediction. These permutations included random shuffling of increasing proportions of sites genome-wide, targeted permutations of predefined genomic windows containing selected variants, and permutations of loci in linkage disequilibrium (LD) with those windows. LD partners were identified using pairwise correlation thresholds (see **Supplemental Figure 2.**), and permutations were applied either to the selected window alone or jointly with subsets of LD-linked sites stratified by LD strength (**Supplemental Figure 3.**). This framework enables systematic evaluation of how predictive performance responds to disruption of adaptive loci, correlated hitchhiking regions, and broader neutral genetic structure, providing an interpretable link between model predictions, genetic architecture, and simulated evolutionary processes. Feature attribution analyses were performed separately for each simulation regime, allowing us to assess how the recovery of adaptive loci depends on the underlying evolutionary scenario.

Comparing EcoLocator to other GEA methods

To benchmark the ability of EcoLocator-derived feature attribution to identify adaptive loci, we compared SHAP-based variant rankings to two genome–environment association (GEA) approaches: the Weighted-Z Analysis (WZA) (Booker et al., 2024) and the top-candidate method (Yeaman, Hodgins, et al., 2016). These comparisons were performed across both simulation regimes described above, allowing us to evaluate method performance under distinct evolutionary scenarios.

A range of GEA methods exist, including mixed-model and latent factor approaches (Rellstab et al., 2015; Günther and Coop, 2013; Frichot et al., 2013), that evaluate associations between individual SNPs and environmental or population-specific covariates. Because linked markers may share evolutionary histories or reflect the effects of linkage disequilibrium and hitchhiking, adaptive signal is also often considered at the level of broader genomic regions (Gautier, 2015; Booker et al., 2024). SNP-level and region-level summaries therefore capture complementary aspects of local adaptation: the former localize candidate sites, whereas the latter summarize the extent to which signal is clustered across linked loci.

WZA and the top-candidate method are both window-based approaches designed to capture this regional structure in adaptive signal. WZA combines SNP-level association statistics across genomic windows using a weighted Z framework, whereas the top-candidate method tests whether a region contains an excess of highly associated SNPs. As Booker et al. argue, aggregating information across linked sites can reduce noise and take advantage of the elevated linkage disequilibrium expected around loci involved in local adaptation. Consistent with this intuition, Booker et al. (2024) showed that WZA and the top-candidate method performed as well as or better than a range of single-SNP methods, including Kendall’s τ , BayPass, LFMM, and RDA, across both strong and weak local adaptation scenarios.

We therefore focus on WZA and the top-candidate method as representative region-based approaches that provide strong benchmarks for comparison. Comparing SHAP-based attribution from EcoLocator to these methods provides a direct test of whether SNP-level importance scores derived from a trained predictive model can recover adaptive loci while preserving resolution at the level of individual variants.

Empirical p-values from SHAP. SHAP values do not inherently provide a measure of statistical significance, as they represent model-derived feature importance scores rather than hypothesis test statistics. To enable direct comparison with GEA methods, which report p-values for association, we transformed SHAP-based rankings into an empirical p-value framework.

SHAP values were first aggregated across samples by computing the mean absolute SHAP value per SNP. Variants were then ranked from highest to lowest based on this aggregated score. To enable comparison with GEA-derived p-values, we calculated empirical p-values using a rank-based false discovery rate (FDR) framework. At each rank position i , all variants up to that point were treated as significant, and the proportion of false positives (neutral variants) relative to the total number of variants in the set was calculated. This assigns each SNP a p-value corresponding to the expected false discovery rate at its SHAP threshold. Because this approach relies on known neutrality labels,

these p-values represent an empirical, supervised estimate of significance rather than a model-based or permutation-based test.

Weighted-Z Analysis (WZA). WZA aggregates SNP-level signals within genomic regions to test whether a window shows evidence of association with environmental variation. Following Booker et al. (2024), we first computed per-SNP association statistics by measuring the correlation between genotype and environmental values, and obtained a p-value for each SNP. These SNP-level p-values were then transformed into Z-scores and combined within genomic windows using a weighted Z-test, where weights are proportional to expected heterozygosity, such that SNPs with higher heterozygosity contribute more strongly to the window-level statistic. This weighting scheme reflects the greater information content of polymorphic sites with intermediate allele frequencies.

To account for variation in the number of SNPs per window, we applied a polynomial-based correction in which the expected mean and variance of window-level Z-scores were modeled as functions of SNP count. Corrected p-values (Z_pVal) were then obtained under the assumption of normality. In parallel, the top-candidate method identifies windows enriched for highly significant SNPs using a binomial test based on the upper tail of the SNP-level p-value distribution.

Integration across methods. For each simulation replicate, we combined SNP-level SHAP p-values, window-level WZA p-values, and top-candidate p-values into a unified dataset. Window-based statistics were mapped back to individual SNPs using SNP-to-window assignments, allowing direct comparison across methods at the SNP level. To obtain a genome-wide assessment, variant-level results were concatenated across simulation replicates within each genetic architecture and simulation regime.

Performance evaluation. Method performance was evaluated using precision–recall (PR) curves, which are appropriate for imbalanced datasets with relatively few adaptive variants. For each method, p-values were transformed to scores ($-\log_{10}(p)$), such that larger values correspond to stronger evidence of association. Precision and recall were computed across all thresholds, and area under the precision–recall curve (AUC-PR) was used as a summary statistic. The random baseline was defined as the proportion of adaptive variants in the dataset. Performance was evaluated separately within each simulation regime, allowing direct comparison of method behavior under both spatially explicit and Wright–Fisher evolutionary scenarios.

Results

EcoLocator prediction accuracy scales proportionally with strength of local adaptation

To explore the evolutionary scenarios under which EcoLocator is most suitable, we evaluated its prediction accuracy across a range of simulation parameterizations (Figure 2). We ran 100 simulation replicates for each parameter combination in our continuous space simulations, and we ran

30 replicates for the discrete space simulations, following Booker et al. (2024). We summarized the distribution of local adaptation scores calculated from each replicate within each regime (see **Supplemental Figure 4.**). Under restricted dispersal ($d = 0.1$) and a smaller standard deviation of the fitness function (stronger selection, $\sigma_K=0.5$), simulated populations became strongly locally adapted to their environments ($LA < 0.1$) after 10,000 generations. EcoLocator performed best when dispersal was constrained ($d = 0.1$, Lon. $R^2=0.968$, Lat. $R^2=0.967$, Env. $R^2=0.769$) (ref. Figure 2, panel a.). As dispersal increased, EcoLocator’s ability to predict both geographic location and environmental covariates declined ($d=0.5$, Lon. $R^2=0.815$, Lat. $R^2=0.814$, Env. $R^2=0.336$) (ref. Figure 2, panel a.). This pattern is consistent with increased gene flow eroding local adaptation and thereby reducing the correspondence between environmental and genetic variation. Location prediction accuracy similarly declining with increasing dispersal likely reflects reduced isolation by distance and weaker population structure in high-dispersal populations.

Decreasing selection strength, implemented through increasing the standard deviation of the fitness function, also reduced EcoLocator’s accuracy for environmental covariate prediction. Under weaker selection, intermediate genotypes are penalized less strongly, resulting in reduced local adaptation and a diminished genetic signal of environmental variation. In contrast to dispersal, varying selection strength did not significantly affect location prediction ($\sigma_K=0.3$, Lon. $R^2=0.917$, Lat. $R^2=0.916$; $\sigma_K=0.5$, Lon. $R^2=0.911$, Lat. $R^2=0.911$; $\sigma_K=0.7$, Lon. $R^2=0.906$, Lat. $R^2=0.907$), which is expected given that dispersal was held constant ($d = 0.1$) across all selection regimes, maintaining a consistent spatial population structure across these simulations. The strength of selection does, however, impact prediction accuracy of the environmental variable ($\sigma_K=0.3$, Env. $R^2=0.609$, $\sigma_K=0.5$, Env. $R^2=0.561$, $\sigma_K=0.7$, Env. $R^2=0.421$)

Varying genetic architecture produced only a slight (negligible) difference in local adaptation (ΔLA between GA1 and GA3 = 0.0008), despite the higher number of QTL mutations arising under genetic architecture 3 relative to genetic architecture 1 (Non-neutral variants in GA1=25, GA3=35). Biologically, varying genetic architecture allows us to explore differences in how adaptive effects are distributed across loci, ranging from architectures dominated by fewer, larger-effect QTL to those involving many smaller-effect and potentially redundant loci contributing to similar phenotypic outcomes. Across architectures, EcoLocator’s location prediction accuracy remained relatively unchanged (GA1 - Lon. $R^2=0.967$, Lat. $R^2=0.967$; GA2 - Lon. $R^2=0.967$, Lat. $R^2=0.967$; GA3 - Lon. $R^2=0.968$, Lat. $R^2=0.968$), and environmental covariate prediction accuracy was largely similar (GA1 - Env. $R^2=0.759$; GA2 - Env. $R^2=0.771$; GA3 - Env. $R^2=0.787$), although predictions under genetic architecture 1 exhibited more extreme outliers and genetic architecture 3 had the highest accuracy, likely due to more non-neutral variants being present. These results suggest that EcoLocator’s performance is robust to differences in genetic architecture within the range explored here, while leaving open questions about how effect size distributions and locus redundancy shape model sensitivity under more complex or realistic adaptive scenarios, which we return to in the Discussion.

EcoLocator also performed strongly under the discrete-space, Wright–Fisher simulation regime (Figure 2, panel b). Across 30 simulation replicates, EcoLocator accurately predicted both deme coordinates and the environmental optimum assigned to each deme (Lon. $R^2=0.986$, Lat. $R^2=0.985$,

Env. $R^2=0.916$). This high prediction accuracy is consistent with the structure of the Wright–Fisher simulations, in which migration is restricted among neighboring demes and local adaptation is driven by directional selection on a small number of causal loci. Together, these features generate strong spatial genetic structure and localized genotype–environment associations. This creates a more clearly structured prediction problem than the continuous-space simulations, because the adaptive signal is concentrated in a small number of causal loci rather than distributed more diffusely across the genome. These results demonstrate how well EcoLocator can recover geographic and environmental signal when population structure and genotype–environment associations are strong, while also highlighting how differences in demographic structure, selection model, and genetic architecture may shape prediction difficulty.

EcoLocator Performance Under Various Simulation Parameterizations

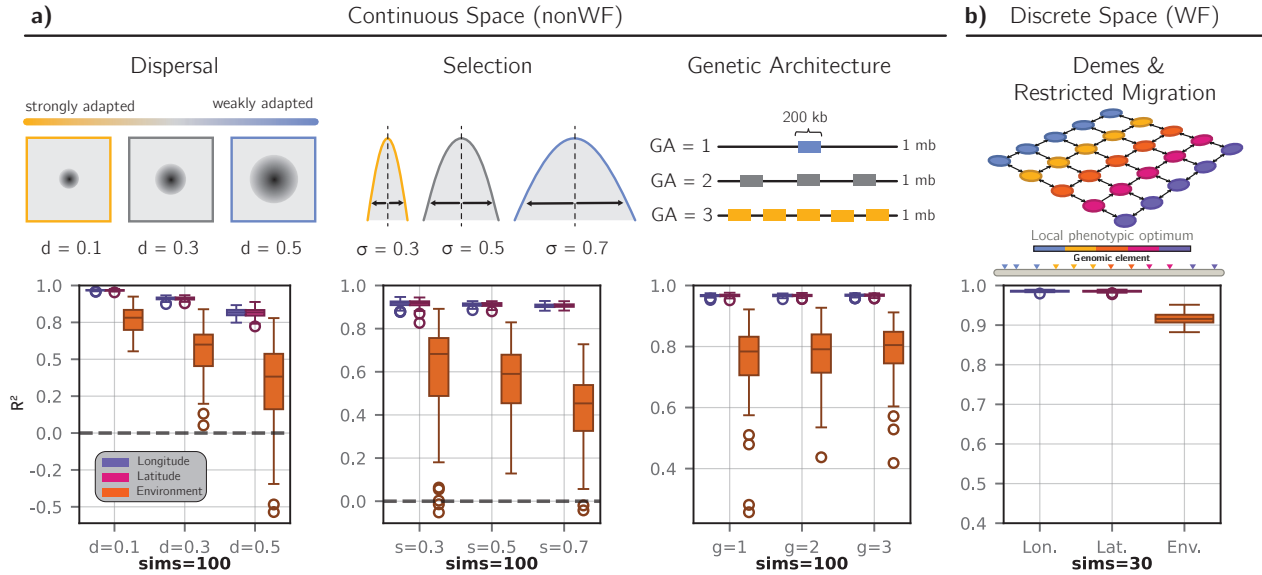


Figure 2: Effects of parameter variation on EcoLocator performance. **Panel a)** shows the continuous-space, non-Wright-Fisher simulation parameterizations in the top row, with prediction accuracy for each respective parameter shown as boxplots in the bottom row. We varied dispersal ($d = 0.1, 0.3, 0.5$), selection strength ($\sigma_K = 0.3, 0.5, 0.7$), and genetic architecture ($GA = 1, 2, 3$). Yellow indicates parameter regimes expected to produce stronger local adaptation, including limited dispersal, stronger selection, and more distributed QTL architectures, whereas blue indicates parameter regimes expected to produce weaker local adaptation, including higher dispersal, weaker selection, and more restricted genetic architectures. Boxplots show prediction accuracy, measured as R^2 using scikit-learn, for longitude, latitude, and environmental targets across the varied simulation parameters. Each continuous-space simulation parameter regime was replicated 100 times. **Panel b)** shows the discrete-space, Wright-Fisher simulation regime in the top row, with corresponding prediction accuracy shown in the bottom row. The Wright-Fisher simulations were held under a fixed parameterization and replicated 30 times, providing a complementary comparison to the continuous-space parameter sweeps. Boxplots summarize EcoLocator prediction accuracy across geographic and environmental targets in this discrete-deme framework.

Feature attribution methods reveal adaptive loci in silico

To evaluate whether feature attribution identifies adaptive loci, we summarized SHAP values across all simulation replicates under strong local adaptation ($d = 0.1$, $\sigma_K = 0.5$) and compared attribution scores between non-neutral variants and neutral variants, including sites in linkage disequilibrium. Across all continuous-space genetic architectures, non-neutral variants consistently received higher SHAP values than neutral sites, indicating that EcoLocator preferentially attributes predictive signal to loci under selection (Figure 3, panels a–c). This separation was evident both in individual simulation examples and in distributions aggregated across all replicates.

Under genetic architecture 1 (Figure 3, panel a), non-neutral variants exhibited higher mean attribution values ($\mu = 0.390$) relative to neutral and LD sites combined ($\mu = 0.292$), with this difference highly significant ($p = 2.21 \times 10^{-4}$). A similar pattern was observed under genetic architecture 2 (Figure 3, panel b), where non-neutral variants again showed elevated SHAP values ($\mu = 0.403$) compared to neutral and LD variants ($\mu = 0.303$; $p = 5.85 \times 10^{-6}$). Despite differences in the number of adaptive loci, genetic architecture 3 (Figure 3, panel c) also showed enrichment of SHAP signal at non-neutral variants ($\mu = 0.394$) relative to neutral and LD sites ($\mu = 0.319$; $p = 8.79 \times 10^{-4}$).

Although effect sizes were modest (Cohen’s $d \approx 0.14$ – 0.18 across architectures), the consistent shift in attribution distributions indicates that adaptive loci are preferentially weighted by the model. Notably, the magnitude of separation between non-neutral and neutral variants was similar across genetic architectures, despite differences in the total number of causal loci and the extent of linkage disequilibrium. This suggests that feature attribution performance is robust to variation in genetic architecture within the range explored here.

We observed stronger separation between neutral and non-neutral sites in the discrete-space Wright–Fisher simulations (Figure 3, panel d), where local adaptation was driven by 12 causal loci. Non-neutral variants had substantially higher SHAP values than neutral variants, with a median attribution value of 0.326 for non-neutral sites compared to 0.132 for neutral sites. This difference was highly significant ($p = 8.17 \times 10^{-9}$), indicating that EcoLocator concentrated attribution signal at causal loci in this discrete-deme simulation regime.

These results show that SHAP-based feature attribution can highlight loci contributing to local adaptation, while also capturing signal from linked regions. The stronger enrichment observed in the Wright–Fisher simulations is consistent with the structure of that regime, where adaptive signal is concentrated in a small number of causal loci. In contrast, the continuous-space simulations likely represent a more diffuse attribution problem, where adaptive signal is distributed across genetic architectures and partially shared with linked regions through hitchhiking or spatially structured linkage disequilibrium.

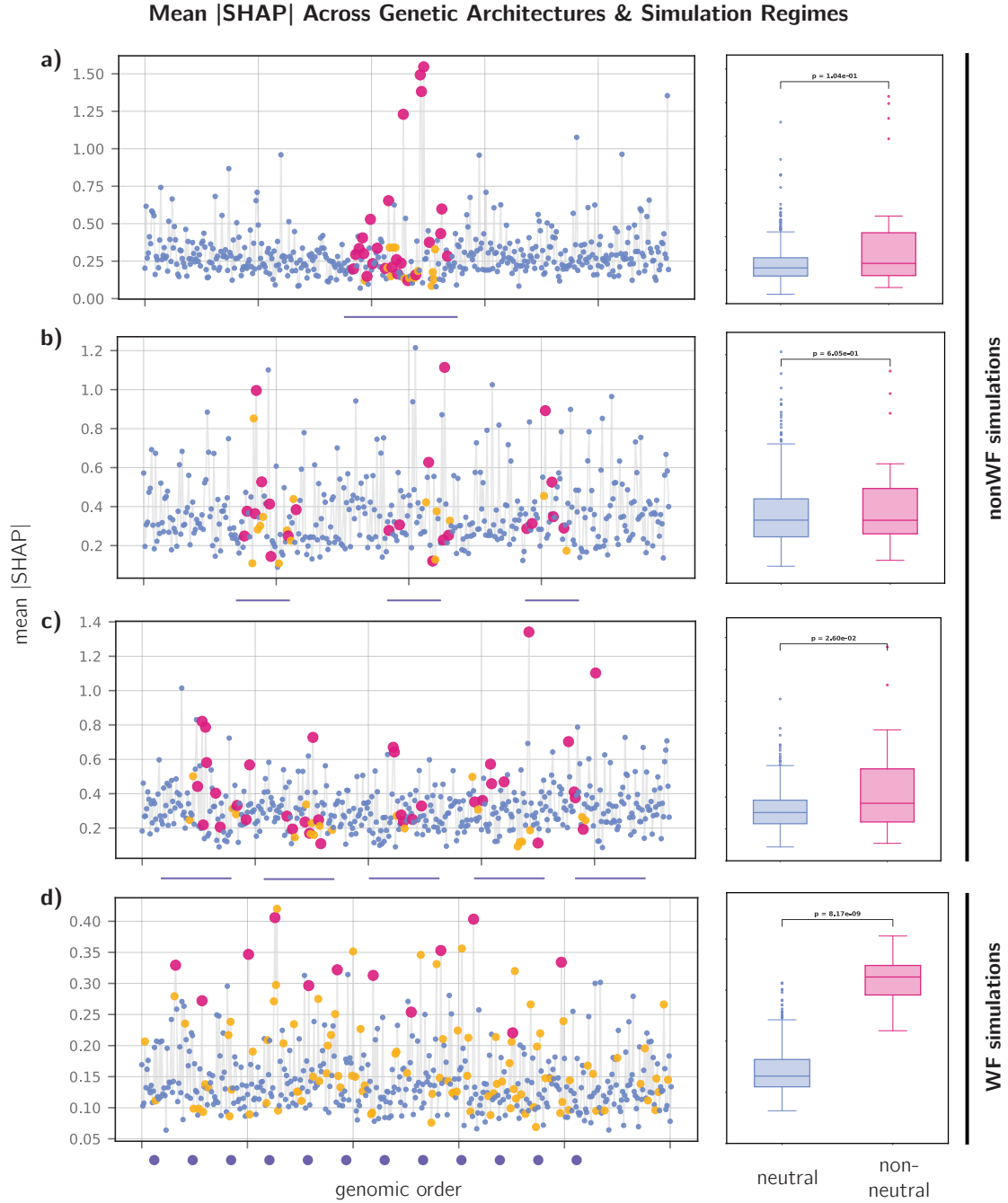


Figure 3: Feature attribution across genetic architectures. Each panel on the left shows mean absolute SHAP scores across the simulated genomic element for neutral sites (blue), non-neutral sites (pink), and sites in linkage disequilibrium with non-neutral variants (yellow). **Panel a)** represents simulations under genetic architecture 1, **Panel b)** represents simulations under genetic architecture 2, and **Panel c)** represents simulations under genetic architecture 3. Across panels a, b, and c, dispersal was held constant at ($d = 0.1$), and the standard deviation of the fitness function was held constant at ($\sigma_K = 0.5$). **Panel d)** shows SHAP attributions for the Wright-Fisher, discrete-space simulation regime, where 12 large-effect causal loci were simulated.

Comparison of EcoLocator to other GEA methods

A notable capability of machine learning approaches like EcoLocator is that, in this context, feature attribution methods provide locus-level importance scores that can be interpreted analogously to genome-environment association statistics. This allows direct comparison between model-derived attribution scores and traditional GEA approaches for identifying adaptive loci. Using SHAP values computed from trained EcoLocator models, we ranked variants according to their contribution to environmental prediction and compared their performance to Weighted-Z Analysis (WZA) (Booker et al., 2024) and the top-candidate method (Yeaman, Hodgins, et al., 2016).

To quantify detection accuracy, we calculated empirical false discovery rate (FDR) p-values from SHAP rankings and evaluated performance using precision-recall curves (Figure 4). The random baseline represents the AUC-PR expected if variants were ranked at random, given the proportion of adaptive variants in each simulation regime. For each simulation regime, the precision-recall curves were constructed by pooling SNPs across all simulation replicates, so the resulting AUC-PR values reflect global discriminatory performance. We also calculated replicate-level AUC-PR values, shown as violin plots, where each point represents one simulation replicate and the horizontal bar indicates the mean. Thus, the pooled AUC-PR values summarize overall variant-level discrimination across all replicates, whereas the violin plots summarize replicate-to-replicate variation in method performance.

Across all continuous-space genetic architectures, SHAP-based attribution consistently recovered a higher proportion of true adaptive variants than either comparison method in the pooled precision-recall analysis. Under genetic architecture 1, SHAP achieved a pooled AUC-PR of 0.173, substantially exceeding WZA (0.079) and the top-candidate approach (0.078), and outperforming the random expectation (0.043). Under genetic architecture 2, SHAP again showed improved performance (pooled AUC-PR = 0.187) relative to WZA (0.065) and the top-candidate method (0.063), with a random baseline of 0.051. Similarly, under genetic architecture 3, SHAP yielded a pooled AUC-PR of 0.179, compared to 0.082 for WZA and 0.079 for the top-candidate approach (random = 0.071).

The replicate-level violin plots showed the same overall ranking. Under genetic architecture 1, EcoLocator SHAP had the highest mean AUC-PR (mean AUC-PR = 0.208), followed by the top-candidate method (mean AUC-PR = 0.128) and WZA (mean AUC-PR = 0.121). Under genetic architecture 2, EcoLocator×SHAP again showed the highest mean performance (mean AUC-PR = 0.221), followed by the top-candidate method (mean AUC-PR = 0.103) and WZA (mean AUC-PR = 0.100). Under genetic architecture 3, EcoLocator×SHAP maintained the highest mean AUC-PR (mean AUC-PR = 0.205), followed by WZA (mean AUC-PR = 0.113) and the top-candidate method (mean AUC-PR = 0.105).

Performance differences were most pronounced at low recall values, where SHAP maintained substantially higher precision, indicating that the highest-ranked loci were enriched for true causal variants. In contrast, WZA and the top-candidate method showed rapid declines in precision, approaching random expectations across much of the recall range. Notably, SHAP maintained improved performance across all genetic architectures, despite differences in the number and genomic

distribution of adaptive loci.

In the discrete-space Wright–Fisher simulations, *EcoLocator*×SHAP showed particularly strong recovery of adaptive loci, achieving a pooled AUC-PR of 0.843 relative to a random baseline of 0.0149 (Figure 4, panel d). WZA also performed well in this regime (pooled AUC-PR = 0.498), whereas the top-candidate method showed substantially lower recovery (pooled AUC-PR = 0.071). Replicate-level AUC-PR values followed the same ordering, with *EcoLocator*×SHAP showing the highest mean performance (mean AUC-PR = 0.906), followed by WZA (mean AUC-PR = 0.619) and the top-candidate method (mean AUC-PR = 0.102).

The higher performance of both *EcoLocator*×SHAP and WZA in the Wright–Fisher simulations is consistent with the way local adaptation was modeled: adaptive signal was concentrated in a smaller number of causal loci with strong, localized genotype–environment associations. These results suggest that feature attribution from *EcoLocator* can be especially effective when adaptive variants produce clear, localized genomic signal, while performance in continuous-space simulations may be more limited when adaptive signal is weaker, more diffuse, or embedded in more complex spatial structure.

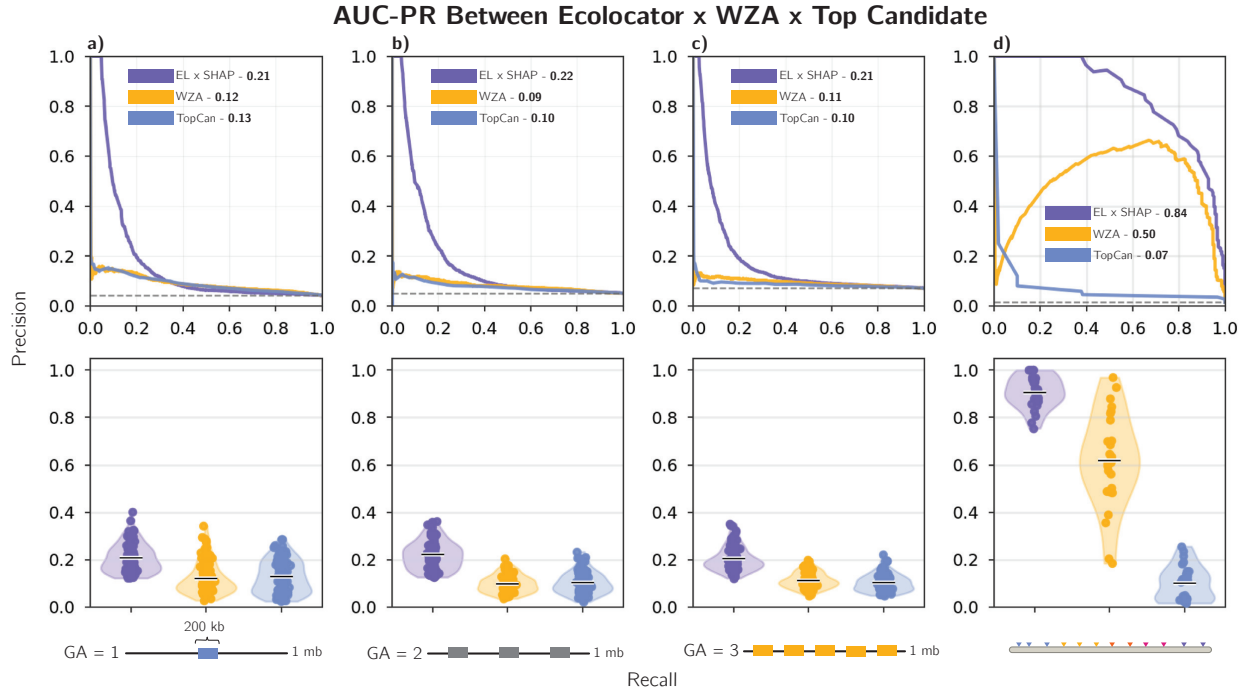


Figure 4: AUC-PR curve comparison between SHAP, WZA, and top-candidate methods. Each column represents a different simulation regime. **Panel a)** represents genetic architecture 1, **Panel b)** represents genetic architecture 2, and **Panel c)** represents genetic architecture 3. **Panel d)** represents the Wright-Fisher, discrete-space simulation regime. The top row shows precision-recall curves constructed by pooling SNPs across all simulation replicates within each regime; the area under each pooled curve reflects global discriminatory performance and is not directly comparable to the per-replicate means shown below. The bottom row shows violin plots of per-replicate AUC-PR values, where each point represents one simulation replicate and the horizontal bar indicates the mean; this view captures replicate-to-replicate variability in performance. In all panels, purple indicates EcoLocator×SHAP, yellow indicates WZA, and blue indicates the top-candidate method.

Case study: Predicting suitable environments for coastal Douglas Fir using EcoLocator

Simulations provide a controlled setting to benchmark EcoLocator, but the ultimate goal is to evaluate its performance in natural systems where demography, gene flow, selection, and management history interact in more complex ways. To this end, we applied EcoLocator to coastal Douglas-fir (*Pseudotsuga menziesii* var. *menziesii*), a keystone conifer of western North American forests that has been the focus of forest genetics research for over a century (J. Bradley St. Clair, Glenn T. Howe, and Kling, 2020). Coastal Douglas-fir spans steep gradients in temperature, precipitation, elevation, and summer drought, and a century of provenance trials and seed-transfer research have shown that trees show maladaptive responses (e.g., slow growth, poor form, disease susceptibility)

when moved beyond their local source climate (J. Bradley St Clair, Mandel, and Vance-Borland, 2005; J. B. St Clair and G. T. Howe, 2007). This pattern is well-documented in seedling and tree provenance trials, which consistently show home-environment advantage and genotype–environment interactions along climatic gradients related primarily to winter cold, summer drought, and summer heat accumulation.

To minimize maladaptation in Douglas-fir, seed zones were developed to define geographic limits within which seedlings can be moved while preserving local adaptation and minimizing maladaptive responses. Douglas-fir seed zones were formalized for U.S. Forest Service forest lands in the 1940s (Kummel, Rindt, and Munger, 1944), and they were later expanded to include state and private lands in the 1960s and 1970s (USDA Forest Service, 1973). Modern seed transfer zones (e.g., Randall, 1996; John Bradley St. Clair et al., 2022) incorporate geographic and climatic factors, as well as experimental evidence from reciprocal transplant trials. This well-developed biological and management framework makes Douglas-fir a particularly powerful empirical system for evaluating EcoLocator.

Data and model application

Our empirical dataset consisted of 348 unrelated coastal Douglas-fir individuals sampled across Oregon and Washington. The maternal parents of these trees were woodsrun (unimproved) ‘plus’ trees, identified through field observations as well-adapted to their local environment and exceptional for diameter, height, and stem form. Since Douglas-fir exhibits low self-fertility, seeds derived from maternal parents are likely pollinated by multiple nearby pollen donors, producing half-sib families instead of full-sibs or selfs. Given the limited movement of Douglas-fir pollen, paternal contributions are expected to be restricted to nearby trees (<400 m), making the maternal source environment a reasonable proxy for the source of both parents.

Leaf-derived DNA from individual trees was genotyped using the 58K Douglas-fir Axiom Affymetrix SNP array (Glenn T. Howe et al., 2020), yielding 18,732 SNPs after filtering. The Douglas-fir genome is large and incompletely assembled (~16 Gb; Neale et al., 2017), so genomic positions of array SNPs are unknown, precluding chromosome- or window-level analyses.

To represent the climate conditions of maternal trees, we chose 1970–2000 baseline climate normals modeled by ClimateNA (Mahony et al., 2022) using source latitude, longitude, and elevation. We focused on three climate variables shown to be significantly associated with Douglas-fir growth and local adaptation (J. Bradley St Clair, Mandel, and Vance-Borland, 2005): mean cold month temperature (MCMT), which shows the strongest associations with the timing of summer budset and annual height increment; and summer heat-moisture index (SHM) and mean summer precipitation (MSP), both of which show strongest associations to the timing of spring bud break and the magnitude of stem taper. These variables capture the primary axes of climate variation across the species’ range and correspond directly to the selection pressures (e.g., winter cold temperatures; aridity) that are managed and mitigated by USFS seed zones.

Prior to model training, we applied standard preprocessing to ensure comparability across genomic and environmental inputs, as detailed in the preprocessing section. These steps minimize

technical artifacts and place all predictors and response variables on compatible scales for joint prediction. Following these preprocessing steps, we trained EcoLocator on this dataset using leave-one-out cross-validation, enabling direct comparison between each individual's predicted and known geographic coordinates and climate values. Geographic coordinates and environmental covariates were centered and scaled prior to model training.

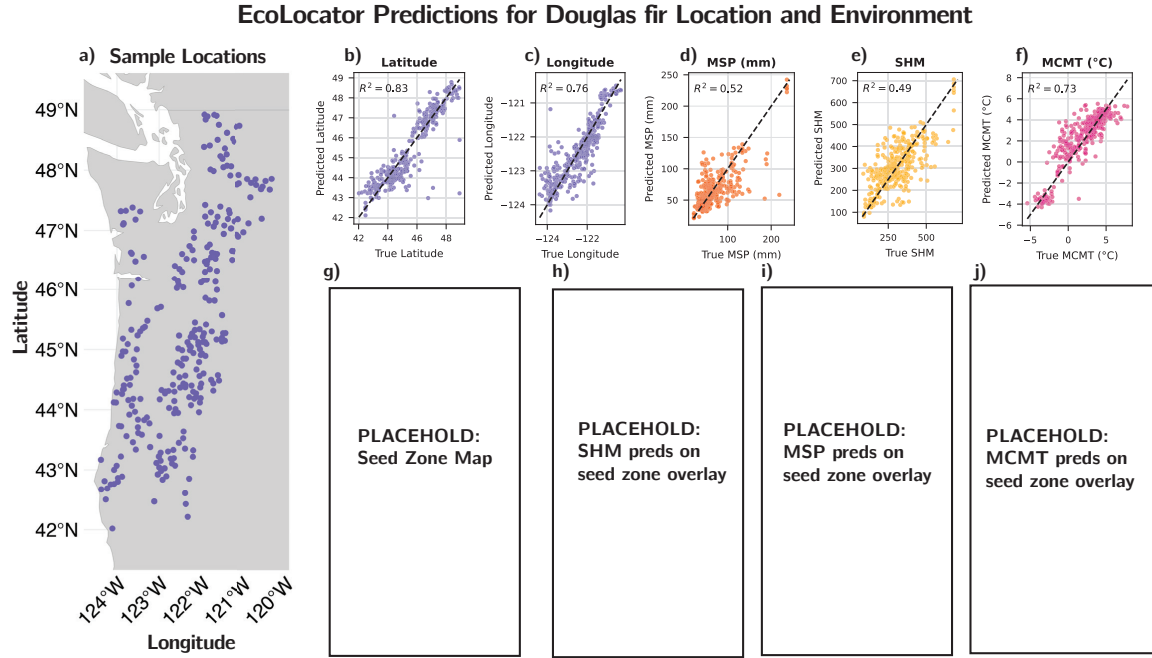


Figure 5: Douglas fir case study results.

Prediction accuracy

EcoLocator predicted source latitudes and longitudes with R^2 values of 0.84 and 0.76 respectively ((Figure 4), Supplemental Table 1.). Combined latitude and longitude for each sample gave predicted geographic origins that showed a median error of 51.6km (see **Supplemental Figure 6.**). Minimum errors were as low as 1.2km, while maximum errors were as high as 646km. Errors greater than 200km appeared as unusual outliers and were not associated with obvious geographic features (e.g. edge of range; clustered locations). Label errors or failed predictions are likely underlying sources for these unusually large errors.

We evaluated climate predictions using two approaches: using direct climate predictions from Ecolocator; or indirect climate predictions from Ecolocator geographic predictions combined with climate values estimated from ClimateNA. Direct estimates from Ecolocator showed higher accuracy than translating locations to climate via ClimateNA variables (ref. **Supplemental Table 1.**), indicating Ecolocator's ability to identify climate-related covariance in the geographic-genomic

matrix, and ability to use this covariance to improve local climate estimates. Among the three modeled climate variables, Mean Cold Month Temperature (MCMT) showed the highest predicted R^2 of 0.73, with May to September Precipitation (MSP) ($R^2=0.52$) and summer heart:moisture index (SHM) ($R^2=0.49$) showing lower and comparable prediction accuracies. These correlations are consistent with climate patterns in the Pacific Northwestern U.S., as monthly minimum temperatures are vastly easier to model than monthly minimum temperatures due to complex interactions between topography and aspect with local precipitation and aridity (Frankson et al., 2022).

Validation against USFS seed zones

To assess whether EcoLocator’s prediction errors are biologically and practically acceptable, we compared EcoLocator-predicted values to climate ranges within USFS seed transfer zones. In the Pacific west, ‘first-generation’ seed transfer zones (USDA Forest Service, 1973) were narrowly defined to restrict movement and minimize maladaptive responses to winter cold temperatures. ‘Second-generation’ zones (Randall, 1996; Randall and Berrang, 2002) were subsequently developed after common garden experiments revealed that Douglas-fir populations show broader climate tolerance than are reflected in first-generation zones. For our comparison, we compiled spatial layers for first-generation Oregon and Washington seed zones, along with climate summaries for seed zones divided into 1000 ft elevation bands, rather than the original 500-foot elevation bands (Shalev et al., 2026). This allows us to evaluate results in the context of first-generation seed zone boundaries (which remain widely used in forest management), while making comparisons to climate transfers that are closer to the experimentally-supported updated second-generation zones. For each individual tree, we compared Ecolocator climate predictions to known seed zone climates using two approaches: (a) using the map-based seed zone as fixed constraint; and (b) using individual seed zone climate ranges to define maximum climate transfers (focal point-based).

In the first comparison, Ecolocator climate predictions were evaluated for whether they fell within a percentile-defined range for each tree’s source seed zone and elevation band. Specifically, we compared predicted climate values to the 100th percentile range (defined by the minimum and maximum values) and the 98th percentile range (defined by the 1st and 99th percentiles). This map-based approach treats zones as fixed, and it can lead to “non-matches” due to the arbitrary nature of seed zone boundaries, even when predictions are near-perfect. Using this approach, we find that 66.1% of the MCMT predictions fall within seed zone–elevation band 100th percentile ranges, and that the number of matches drop to 49.7% of predictions when using the 98th percentile range. The proportion of matches is higher for MSP (86.2% for the 100th percentile range) and SHM (87.6% for the 100th percentile range)(Table 1). These values are all larger than similar estimates using ClimateNA predictions from Ecolocator geographic predictions (Table 1).

In the second comparison, Ecolocator MCMT predictions were evaluated relative to the range of climate variation within the source seed zone and the range of climate variation between known and predicted values. For example, if a source seed zone–elevation band shows a MCMT 100th percentile range of 2°C, an Ecolocator MCMT prediction within 2°C of the true value fall within the range of transfers allowed within seed zones. This type of ‘focal point’ match considers the magnitude of

Table 1: Comparison of EcoLocator and Location+ClimateNA predictions across climate variables.

Factor	EcoLocator		Locator+ClimateNA	
	100%ile range	98%ile range	100%ile range	98%ile range
MCMT	66.1%	49.7%	65.2%	48.9%
MSP	86.2%	75.9%	74.1%	57.5%
SHM	87.6%	74.7%	71.6%	55.5%

climate difference between the known and predicted values at a site, and the climate range of that specific seed zone, rather than arbitrary geographic boundaries. Using this approach, we find that 93.1% of the MCMT predictions show lower distances from the true value than the 100th percentile range of the source seed zones, and that the number of matches drop to 87% of predictions when using the 98th percentile range (Table 2). Predictions for MAP and SHM showed higher accuracies, with 98.6% and 97.7% of estimates falling within the 100th percentile range of the source seed zone. As was observed with predictions within seed zones, Ecolocator showed higher prediction accuracy than estimates using ClimateNA predictions from Ecolocator geographic predictions (Table 2).

Table 2: Comparison of EcoLocator and Location+ClimateNA predictions across climate variables.

Factor	EcoLocator		Locator+ClimateNA	
	100%ile range	98%ile range	100%ile range	98%ile range
MCMT	93.1%	87.1%	89.9%	75.9%
MSP	98.6%	93.1%	94.5%	85.3%
SHM	97.7%	92.8%	91.9%	83.6%

Prediction success was not uniform across seed zones (REF FIG SEED ZONE). In our sample, trees from coastal seed zones in Oregon and Washington showed higher accuracy predictions for climate variables (MCMT, MSP, SHM) than samples from the Cascade Range. Seed zones with highest prediction errors (e.g., 90, 473, 493, 501, 652, and 661; REF MAP IN FIG 5) are primarily located near the crest of the Cascade Range, which is a natural boundary that divides watersheds and climates, and a region with higher climatic complexity than coastal regions. Coincidentally, these specific zones also have small sample sizes (1 – 2 trees per seed zone), and this may account for their unusually high mean error.

These patterns are therefore consistent with a combination of ancestry, topographic complexity, and spatial sampling limitations rather than a simple, systematic failure of the model. Increased sampling across topographically complex regions and across the coastal–interior transition zone would provide a useful future test of whether these Cascade-crest anomalies persist, are associated with interior-variety introgression, or reflect insufficient local sampling.

These results demonstrate that EcoLocator can predict the climate of origin of individual genotypes with accuracy that is both statistically meaningful and relevant for seed transfer management. They further illustrate the value of seed zone benchmarking as a biologically grounded validation framework: predictions that fall within a seed zone climate envelope are not just numerically ac-

curate, but they are close enough that forest managers would consider it an acceptable transfer source.

Discussion

Joint prediction of geography and climate reveals spatial structure and local adaptation

It is well established that multilocus genomic variation encodes both neutral spatial structure and signals of environmental adaptation. Genetic principal components mirror geography (Novembre, Johnson, et al., 2008), allele frequencies covary with environmental gradients under local selection (Coop et al., 2010; Savolainen, Lascoux, and Merilä, 2013), and isolation by environment can generate genetic structure independent of geographic distance (Wang and Bradburd, 2014). Continuous spatial structure leaves particularly strong and pervasive signatures in genomic data, influencing diversity statistics, demographic inference, and genotype–environment associations in ways that discrete-deme models fail to capture (C. J. Battey, Peter L. Ralph, and Andrew D. Kern, 2020; Chevy et al., 2025). C. Battey, Peter L Ralph, and Andrew D Kern (2020) demonstrated that these spatial signatures are rich enough to predict the geographic origin of individuals from genotypes alone using deep neural networks, achieving high accuracy across diverse taxa. EcoLocator extends this framework by jointly predicting geographic location and climate of origin, unifying spatial and environmental inference in a single supervised model. This allows us to ask not only where an individual comes from, but what environmental conditions it is adapted to, and how the recovery of each signal depends on evolutionary parameters. Our simulation results show that prediction accuracy scales with the strength of local adaptation: under restricted dispersal and strong stabilizing selection, the model accurately recovers both the spatial position and the environmental conditions experienced by individuals. As dispersal increases or selection weakens, environmental prediction degrades while geographic prediction is affected primarily by dispersal. This decoupling is consistent with the expectation that these two signals have distinct evolutionary origins: geographic prediction depends on isolation-by-distance patterns driven by limited gene flow, whereas environmental prediction requires allele frequency clines that track ecological gradients—a signal that is eroded when gene flow overwhelms local selection.

The Wright–Fisher results illustrate the upper end of EcoLocator performance under strong, structured genotype–environment associations. In contrast to the continuous-space simulations, where individuals vary continuously across space and local adaptation emerges under stabilizing selection on a quantitative trait, the Wright–Fisher simulations impose spatial structure through discrete demes and directional selection on a small number of causal loci. EcoLocator performed particularly well in this setting (Lon. $R^2=0.986$, Lat. $R^2=0.985$, Env. $R^2=0.916$), suggesting that the model readily recovers geographic and environmental signal when spatial structure is strong and genotype–environment associations are concentrated in the genome. This result is consistent with the broader pattern observed across simulations: EcoLocator performs best when evolutionary processes generate a clear correspondence between genomic variation, geographic position, and environmental

conditions. At the same time, the contrast between the Wright–Fisher and continuous-space simulations highlights that prediction difficulty depends on the structure of the evolutionary problem. Discrete demes, neighbor-limited migration, and a small number of causal loci create a more clearly structured prediction task, whereas continuous space, diffuse genetic architectures, high dispersal, or weak selection can weaken the signals that EcoLocator is designed to learn.

The Douglas-fir case study extends these findings to a natural system where the ground truth is not fully known. EcoLocator’s predictions of geographic origin and climate variables in coastal Douglas-fir are consistent with the species’ well-documented patterns of local adaptation along steep latitudinal and elevational gradients in western North America (Aitken and Whitlock, 2013). That the model performs well in this system (Lon. $R^2=0.76$, Lat. $R^2=0.83$, MCMT $R^2=0.73$, MSP $R^2=0.52$, SHM $R^2=0.49$)—despite a documented low population differentiation (Krutovsky et al., 2009), the absence of a contiguous reference genome (Neale et al., 2017), and a relatively modest number of SNPs—suggests that the joint prediction framework captures biologically meaningful signal even with incomplete genomic resources.

Strengths and limitations

A key strength of EcoLocator is its multitask design, which simultaneously models geographic and environmental variation. This joint framework offers advantages over methods that treat spatial and environmental signals separately: by sharing information across prediction targets during training, the model can leverage correlations between spatial structure and environmental gradients while also learning to distinguish the two. This design also sidesteps a well-documented pitfall of conventional GEA methods. When population structure aligns with environmental gradients—as it often does in nature—methods that correct for structure can inadvertently remove true adaptive signal along with neutral confounders (Lotterhos, 2019; Booker et al., 2024). Because EcoLocator models geography and environment jointly rather than regressing one out of the other, it avoids this over-correction while still accounting for spatial structure through the geographic prediction task. Furthermore, the multitask framework is naturally suited to systems where adaptation occurs along multiple environmental axes. Booker et al. (2024) noted that window-based GEA methods can lose power when adaptation involves semi-independent environmental gradients, because the LD patterns around causal alleles may differ across gradients. EcoLocator’s ability to predict multiple environmental covariates simultaneously may help retain signal in such cases, though this remains to be tested explicitly. The approach is agnostic to genetic architecture, performing comparably across simulations with clustered and distributed QTL arrangements, and it requires no prior specification of candidate loci or assumptions about the genetic basis of adaptation.

However, several limitations should be noted. First, EcoLocator is a supervised method that requires georeferenced samples paired with environmental data for training. Its predictions are therefore bounded by the geographic and environmental range of the training set, and extrapolation to novel environments or unsampled regions should be interpreted cautiously. Second, like all machine learning approaches, the model learns correlations rather than causal mechanisms. High prediction accuracy for an environmental variable does not by itself demonstrate that selection on

that variable has shaped genetic variation. As Booker et al. (2024) emphasize, GEA signals do not constitute a test of local adaptation: correlated environmental predictors, spatially structured neutral variation, and allelic surfing during historical range expansions can all generate allele frequency patterns that mimic adaptive signals (Klopfstein, Currat, and Excoffier, 2006; Novembre and Di Rienzo, 2009). Isolation by distance alone can inflate genotype–environment correlations when environmental gradients are spatially autocorrelated (Meirmans, 2012). EcoLocator’s joint prediction framework does not resolve this fundamental ambiguity; high environmental prediction accuracy could reflect true adaptive signal, demographic history, or both. Third, our simulations use a simplified landscape with a single environmental axis and a modest genome size. Natural systems involve many correlated environmental variables, complex demographic histories, and genomes orders of magnitude larger, all of which may affect model performance in ways not fully captured here. Finally, the resolution of feature attribution is inherently limited by linkage disequilibrium: SHAP values highlight genomic regions rather than individual causal variants, particularly when adaptive loci are in LD with surrounding neutral sites.

Implications for Douglas-fir management and forestry

For natural resource managers, EcoLocator addresses a practical need: identifying the geographic and climatic source of biological materials under historical, current, and projected future climates. Seed transfer guidelines have been used on U.S. Forest Service lands for over a half century, but “off-site” plantings were common prior to the 1940s, and widespread restoration efforts with non-local materials have resulted in well-documented cases of maladaptation. These plantings have the potential to add new knowledge about the growth and fitness of provenances planted in novel environments, but only if their sources can be reliably determined. Similarly, forest tree provenance tests established outside native ranges often identify high-performing genotypes, but lost tree tags or incomplete records can limit breeding advances when the source environments of those genotypes are unknown, making it difficult to identify new germplasm from similar climates. Efforts to combat illegal logging also increasingly rely on DNA-based methods to identify the geographic source of tissues such as wood. Adding climatic source information to these investigations could help discriminate among geographically similar but climatically distinct sources. Finally, traditional reforestation programs based on seed transfer guidelines and common-garden trials are increasingly strained by the pace of climate change (Aitken and Bemmels, 2016), and new tools are needed to manage climate uncertainty. Genomic tools that predict the climate of origin of a given genotype offer a unique strategy for answering these questions, enabling managers to evaluate whether replanting resources are well-matched to the conditions they will encounter at a given site.

Coastal Douglas-fir is a natural test case for this application. Its economic importance, extensive provenance trial data, and strong signals of local adaptation along climate gradients provide both the motivation and the validation framework for genomic prediction tools. By predicting climate of origin alongside geographic location, EcoLocator provides information that is directly relevant to seed transfer decisions: two seed sources from similar geographic locations may differ in their predicted climate of origin, reflecting adaptive differences that would be missed by geography-based

guidelines alone. By comparing EcoLocator predictions to Douglas-fir seed zones, these predictions become interpretable in definable management terms. For example, the original first-generation seed zone–elevation bands in Oregon and Washington span approximately 1.5°C in MCMT across the 98th percentile range, whereas current second-generation seed zones span approximately 2.5°C in MCMT across the same percentile range [CITE]. Our results suggest that EcoLocator predictions often fall within the climate tolerance thresholds that forest managers already use to define locally adapted reforestation material for Douglas-fir. Importantly, this level of accuracy was achieved using a modestly sized dataset of 348 trees and approximately 18,000 SNPs without known genomic coordinates. Although additional sampling is not evaluated here, previous work with Locator suggests that geographic prediction accuracy can improve with increased sampling density and, to a lesser extent, increased genomic sampling (C. Battey, Peter L Ralph, and Andrew D Kern, 2020). Similar improvements may be possible for EcoLocator, particularly in regions where current predictions are limited by sparse sampling or complex environmental gradients.

At the same time, the interpretation of EcoLocator predictions depends on the sampling history and biological structure of the material being analyzed. The Douglas-fir individuals used here were designated as "plus" trees, meaning they were identified through provenance trials or field observations as being particularly well-adapted to their local environments. This selection criterion may strengthen genotype–environment correspondence and aid climate prediction, but it could also overestimate performance relative to less-characterized or more heterogeneous samples. In addition, because genotyping was performed on seeds from monoecious Douglas-fir, seed genotypes served as proxies for the sampled maternal source trees and their origins. However, seed genotypes include both maternal and paternal contributions, and each seed tree may be pollinated by multiple nearby pollen donors, producing half-sib families rather than genetically identical representatives of the maternal tree. Given the limited movement of Douglas-fir pollen, paternal contributions are likely to come primarily from nearby trees, but this remains a source of uncertainty when assigning each genotype to a maternal source environment.

The spatial distribution of prediction error also highlights important considerations for applying EcoLocator in complex landscapes. In this dataset, higher prediction errors were concentrated in parts of the Cascade Range, where several non-mutually exclusive factors may reduce prediction accuracy. First, steep elevational gradients near the Cascade crest generate sharp climate transitions that may be difficult to model, especially with relatively low local sampling density. Second, many seed sources in this dataset predate civilian GPS use, and source locations were recorded using Township–Range–Section estimates. Estimation, rounding, and digitization errors may therefore introduce positional inaccuracies that are amplified in topographically complex terrain. Finally, Douglas-fir populations east of the Cascade crest show evidence of introgression with interior Douglas-fir (*Pseudotsuga menziesii* var. *glauca*) [CITE], and trees with greater interior ancestry may show higher prediction error when evaluated with a model trained primarily on coastal Douglas-fir genomic and climatic data. These patterns suggest that prediction errors are not only statistical residuals, but can also point to biologically meaningful features of the study system, including ancestry transitions, complex topography, and uncertainty in historical source records.

More broadly, the approach is not limited to Douglas-fir or to species with extensive prior char-

acterization. Any system with georeferenced genotypes and associated environmental data can be analyzed with EcoLocator, making it a potentially useful tool for less-studied species where common-garden data are scarce or absent. In these systems, even weak or uneven predictive performance may be informative. Poor environmental prediction may suggest that genetic variation is not strongly tracking the environmental variable being modeled, whereas poor geographic prediction may indicate weak population structure, high gene flow, or limited spatial signal in the sampled genomic data. Conversely, differences in predictive accuracy among environmental variables can help identify which environmental axes are most strongly reflected in genomic variation, providing hypotheses about the selective pressures shaping local adaptation. However, these applications will require careful attention to the provenance, sampling design, and reproductive biology of the focal system. For highly curated datasets, such as provenance trials or plus-tree collections, EcoLocator may provide a strong test of climate-of-origin prediction under conditions where genotype–environment relationships are expected to be pronounced. For less-characterized datasets, predictions may be more sensitive to mixed ancestry, family structure, uncertain source records, or mismatch between sampled tissue and the individual or population whose origin is being inferred. These considerations do not limit the utility of EcoLocator, but they highlight the importance of interpreting predictions within the biological and management context of each application.

Feature attribution as a tool for identifying adaptive loci

SHAP-based feature attribution provides a window into which genomic regions drive EcoLocator’s predictions. Our simulation results demonstrate that non-neutral variants consistently receive higher attribution scores than neutral sites, confirming that the model preferentially weights loci under selection. This signal persists across genetic architectures despite modest effect sizes (Cohen’s $d \approx 0.14$ – 0.18), indicating that SHAP can detect the distributed signature of polygenic adaptation even when individual locus effects are small.

However, the modest effect sizes also highlight an inherent challenge: under polygenic adaptation, the signal at any single locus is subtle, and attribution scores for adaptive and neutral loci overlap substantially. This is consistent with the broader finding that GEA methods are generally underpowered under weak selection, and that detecting most causal loci requires tolerating a substantial false positive rate (Lotterhos and Whitlock, 2015; Booker et al., 2024). EcoLocator’s feature attribution faces the same trade-off: thresholding on SHAP values to identify candidate loci will inevitably involve a balance between sensitivity and specificity that depends on the unknown strength of selection and the genetic architecture of adaptation. Linkage disequilibrium further complicates interpretation, as SHAP values are elevated not only at causal variants but also at linked neutral sites. This is expected, because LD spreads adaptive signal across genomic neighborhoods, but it means that high-attribution regions should be interpreted as candidate windows rather than as individual causal variants. This issue was especially apparent in our initial simulations, where a recombination rate of (10^{-8}) within a 1 Mb genomic element generated extensive LD between the selected window and distant sites (ref **Supplemental Figure 2., panel a.**). Increasing recombination to (10^{-6}) reduced this long-range LD and provided a clearer setting for evaluating whether

SHAP could distinguish causal regions from linked background signal (ref **Supplemental Figure 2., panel b.**).

The permutation-based sensitivity analyses complement SHAP by providing a different lens on the same question. By disrupting specific genomic regions and measuring the resulting loss in prediction accuracy, these analyses show how much predictive information is carried by the selected window, by linked sites, and by genome-wide background variation. Permuting the selected window alone produced only modest changes in prediction performance, whereas jointly permuting the selected window with LD-linked sites caused much larger losses in accuracy (ref **Supplemental Figure 3.**). For example, when the selected window and all LD-linked sites were permuted together, prediction accuracy dropped below zero for all three targets (X ($R^2 = -0.114$), Y ($R^2 = -0.181$), environment ($R^2 = -0.267$)), indicating that the disrupted regions contained most of the signal needed for accurate prediction. In contrast, random genome-wide permutations caused a gradual loss of performance as larger fractions of SNPs were shuffled, with complete randomization producing similarly poor performance (X ($R^2 = -0.507$), Y ($R^2 = -0.426$), environment ($R^2 = -0.425$)). These findings suggest that EcoLocator is not relying on the selected window alone, but on a broader LD-associated haplotypic signal that can obscure the distinction between causal and linked variants.

The SHAP and permutation analyses show that EcoLocator’s predictions are grounded in biologically meaningful genetic variation, but they also highlight an important limitation of feature attribution: when LD is strong, linked neutral sites can mask or share signal with the true causal loci. In this sense, SHAP may be most reliable for identifying candidate genomic regions or haplotypic blocks associated with local adaptation, rather than for pinpointing individual causal mutations. This interpretation is particularly important for empirical systems, where LD can reflect physical linkage, demographic history, selection, population structure, or unobserved ancestry, and where the true causal variants are rarely known.

Notably, SHAP-based attribution from EcoLocator shows greater power to identify adaptive loci than WZA (Booker et al., 2024) and top-candidate approaches in our simulation comparisons (Figure 4). This advantage likely reflects the fact that SHAP attribution operates on a model that has already learned the joint relationship between genotypes, geography, and environment, rather than testing each locus independently against a single environmental predictor. By conditioning on a trained multitask model, SHAP can capture contributions from loci whose individual marginal correlations with environment are weak but whose combined effect is detectable in the context of the full genotype. This shows that supervised prediction followed by post hoc feature attribution may be a powerful complement to traditional GEA approaches, particularly under polygenic adaptation where per-locus signals are subtle.

Future directions

Several extensions of this work merit exploration. First, our simulations assume a purely additive genetic architecture, but epistasis and pleiotropy are pervasive features of quantitative trait variation (Mackay and Anholt, 2024) that can substantially alter the outcome of genome scans for local

adaptation (Jones, Arnold, and Bürger, 2019). Epistatic interactions may generate adaptive allele combinations whose effects are invisible to single-locus attribution methods, while pleiotropic constraints can limit or redirect the response to spatially varying selection (Yeaman, 2022). Extending simulations to include non-additive architectures would clarify whether EcoLocator’s prediction accuracy and feature attribution remain robust when the genotype–phenotype map is more complex. Second, expanding the environmental complexity of simulations—multiple correlated environmental axes, spatially autocorrelated landscapes, and temporally varying selection—would better approximate the conditions under which EcoLocator will be applied in practice. Third, the framework could be extended to predict performance under future climate scenarios by applying trained models to projected environmental surfaces, generating spatially explicit maps of predicted maladaptation that could inform assisted migration planning. Finally, systematic comparison with a broader suite of GEA methods across a wider range of evolutionary scenarios would help delineate the conditions under which EcoLocator’s joint prediction approach offers the greatest advantage over single-task or univariate alternatives.

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¹¹²⁹ **Supplementary Materials**

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